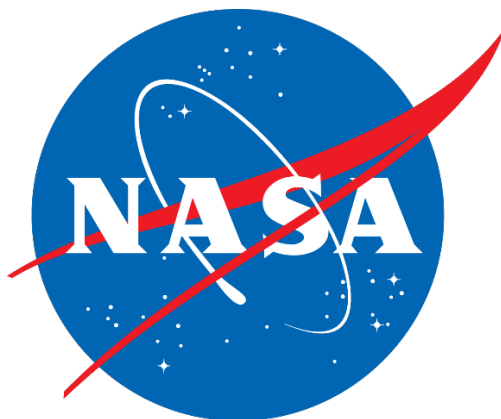




NSSC NCCIPS V3

Draft Performance Work

Statement



Table of Contents

1.0	INTRODUCTION	4
1.1	NCCIPS OVERVIEW	4
1.1.1	Description	4
1.1.2	Services Offered	5
1.1.3	Strategic Objectives	5
1.2	OBJECTIVE	5
2.0	PROGRAM MANAGEMENT	7
2.1	CONTRACTOR EMPLOYEE REQUIREMENTS	8
2.1.1	Contractor Personnel Training	8
2.1.2	Contractor Employee Data	8
2.1.3	Contractor Personnel Access	9
2.1.4	Designation of Position Risk Level	9
2.1.5	Equal Employment Opportunity	9
2.2	RECORDS, DOCUMENTS, AND FORMS MANAGEMENT	9
2.3	REPORTING / INFORMATION REQUESTS	10
2.4	NCCIPS SITE VISITS / INSPECTIONS	10
2.5	PERFORMANCE METRICS REPORTING	11
2.6	INNOVATION AND CONTINUOUS PROCESS IMPROVEMENTS	11
2.7	SECURITY	11
2.7.1	Physical Security	12
2.7.2	Personnel Security Clearances	12
2.7.3	Key Control, Media Access and Protection	12
2.7.4	On-Boarding, Departures and Inter-Office Transfers	13
2.8	SAFETY AND OCCUPATIONAL HEALTH	13
2.8.1	Disaster Recovery and Emergency Situations	15
2.8.2	Occupational Health	16
2.9	RISK MANAGEMENT AND MITIGATION	16
2.10	QUALITY CONTROL	17
2.11	CONTRACTOR FINANCIAL MANAGEMENT	17
2.12	ACQUISITION SUPPORT	18
2.13	PHASE-IN PLAN	20
2.14	NCCIPS CUSTOMER SUPPORT HOURS	20
2.15	CERTIFICATIONS, LICENSES, AND PERMITS	20
2.16	ENVIRONMENTAL REQUIREMENTS	20
2.17	EMERGENCY CONTRACTING	20
2.18	PROPERTY MANAGEMENT	20
2.19	GENERAL MAIL SERVICES	21
2.20	NEW BUSINESS SUPPORT	22
2.21	PRINT AND COPY	22
2.22	WASTE MANAGEMENT	22
2.23	RIGHTS IN DATA	22

2.24	PLACE OF PERFORMANCE	22
2.25	KEY PERSONNEL	23
2.26	CONTINGENCY SUPPORT	24
2.27	PHASE-IN & PHASE-OUT SERVICES	24
3.0	FACILITIES MANAGEMENT	25
3.1	FACILITIES OPERATIONS AND MAINTENANCE	25
3.1.1	Facilities Operations	26
3.1.1.1	Shipping & Receiving	28
3.1.1.2	Custodial and Recycling	28
3.1.1.3	General Materials & Supplies	29
3.1.1.4	Inspections	29
3.1.2	Facilities Maintenance	30
3.1.2.1	General Facility / Grounds Maintenance	31
3.2	ELECTRICAL OPERATIONS	31
3.3	ENGINEERING	32
3.4	REPAIR AND ALTERATION	32
3.5	ENERGY MANAGEMENT	33
3.6	INFORMATION TECHNOLOGY	34

1.0 INTRODUCTION

The National Aeronautics and Space Administration (NASA) Shared Services Center (NSSC) provides data center shared facilities to Customers through the management of the National Center for Critical Information Processing and Storage (NCCIPS) located at the Stennis Space Center (SSC) in Hancock County, Mississippi. This Performance Work Statement (PWS) describes the performance objectives for operating and maintaining the NCCIPS data center facilities and supporting infrastructure. The Contractor shall exercise management and operational control over and retain full responsibility for the performance requirements set forth in this PWS.

For purposes of this document, “Customer” refers to Federal Agencies choosing to utilize co-located data center facility management services offered by NCCIPS. All work effort contained in this PWS is considered CORE SERVICES unless specifically identified as DEMAND which will be performed as delineated in PWS Section 3.3 and PWS Section 3.4 and reimbursed in accordance with PWS Section 2.12. DEMAND activity will be driven by Customer requested project support and/or NASA requested project support.

The requirements and work descriptions included in this PWS have been organized by activity. This grouping is not intended to dictate the Contractor’s organizational structure. It is the Contractor’s responsibility to create the most effective and efficient organization to meet all of the requirements of this PWS.

1.1 NCCIPS OVERVIEW

NCCIPS was a result of a 2005 Congressional initiative to provide secure processing and storage of nationally sensitive, critical, or classified data. The establishment of the NCCIPS data center directly supports Federal government organizations in a shared services model. Secure facility operations, maintenance, management, environmental analysis, energy management, Information Technology (IT), system power, and network integration are the predominant service components of this PWS necessary to support the Customer-owned critical processing and storage equipment and internal NCCIPS facilities support equipment located within the data processing and storage center buildings and throughout the campus. Several Federal Agencies currently utilize the NCCIPS and others are anticipated to transition to the NCCIPS during the period of performance of this PWS.

1.1.1 Description

The NCCIPS is a federal shared services data center facilities campus managed by the NSSC serving federal Customers via interagency agreements. The NCCIPS is located on a 64 acre campus within the boundaries of SSC in southern Mississippi. Leveraging the extensive infrastructure presents Customers significant cost savings/avoidance. Each NCCIPS Customer pays an apportioned share of the actual data center operations and maintenance cost based on each Customer’s footprint in the data center. NCCIPS Customers can estimate their potential annual shared costs by multiplying the number of square feet reserved for the Customer by the established square foot rate. In addition to this service chargeback, Customers pay a monthly metered electric bill for actual power consumption. NCCIPS Customers procure and manage communications circuits directly with their provider(s).

1.1.2 Services Offered

The Tier III level primary services offered to Customers of the NCCIPS are:

- a. Co-location services to NASA, Department of Defense, and other Federal Agencies;
- b. Supporting infrastructure (power, cooling, and fiber cable plant);
- c. Completely redundant electrical distribution from the national grid to the rack;
- d. Commercial power capacity is 32MVA;
- e. Commercial power backed up by 21MW of diesel generator capacity and fifteen 1000KVA Uninterruptible Power Supply (UPS);
- f. Metered electric bill at a kilowatt per hour rate lower than the national average;
- g. 24x7x365 onsite security on a Federal campus;
- h. Onsite and on call maintenance staff in a facility built to withstand significant environmental challenges; and
- i. Historical overall facility uptime of 99.999% and conditioned power uptime to IT equipment of 100%.

1.1.3 Strategic Objectives

The current strategic objectives for the NCCIPS follow:

- a. Develop complete mapping of infrastructure components to improve service management, to streamline operations, and to enhance vulnerability assessments;
- b. Complete assessment of infrastructure standardization opportunities to increase systems efficiency and reduce cost of operations;
- c. Create/review dependency mapping between integrated Customer requirements and NCCIPS infrastructure to improve capacity management;
- d. Implement a discovery and dependency tool to reduce risks and address compliance issues with NCCIPS infrastructure systems; and
- e. Investigate opportunities for network consolidation between Customers identifying risks and associated cost reductions to both Customers and NCCIPS.

1.2 OBJECTIVE

The objective of this acquisition is to acquire the services needed to operate, maintain and manage the NCCIPS facility and campus including equipment maintenance and power management of the SSC telecommunications facility. The NCCIPS campus is a Tier III equivalent data center as defined by the Uptime Institute Tier Classification System. The NCCIPS is a secure facility per Presidential Policy Directive PPD-21 “Critical Infrastructure Security and Resilience” and requires risk management per the most recent edition of “The Risk Management Process for Federal Facilities: An Interagency Security Committee Standard.” This Standard defines the criteria and processes that those responsible for the security of a facility should use to determine its facility security level and provides an integrated, single source of physical security countermeasures for all nonmilitary Federal facilities. The Standard also provides guidance for customization of countermeasures for Federal facilities. This acquisition shall enable the Government to accomplish its Data Center Optimization Initiative (DCOI) objectives established in OMB memorandum M-16-19.

Mission objectives include operating, maintaining and managing a secure 24x7x365 Tier III equivalent shared data center facility infrastructure for critical Federal Agency processing and storage. The Contractor shall provide 24x7x365 critical facility management, operations,

maintenance, IT, system power, and network integration. The effort required is a total solution which supports the data center itself and all ancillary facilities including administrative areas, storage areas, shipping and receiving facilities, and the SSC telecommunications facility.

The physical environment consists of the following:

- a. Thirty-four (34) buildings/structures for a total of 232,600 square feet spread over 64 acres. The main data center facility and associated office space accounts for 147,000 square feet of the total. Not all buildings have power and not all are climate controlled. See Technical Exhibit 1 (TE-1), Facility Support Matrix, for details on the buildings and the corresponding support matrix indicating a mix of support between SSC and NCCIPS;
- b. Electrical power distribution infrastructure and backup power generation;
- c. Heating, Ventilation, and Air Conditioning (HVAC) systems (primarily chilled water);
- d. Data Center raised floor infrastructure support equipment, maintenance and operations for electrical feed controls, generator back-up, high and low voltage switchgear, UPS units, chillers, air handlers, Computer Room Air Conditioning (CRAC) units, Trane Building Automation System (BAS) and Schneider/Square D Supervisory Control and Data Acquisition (SCADA) system for electrical monitoring, and all other equipment integral to Data Center operations;
- e. Diesel fuel farm;
- f. Battery UPS;
- g. Access control systems and security cameras;
- h. Lightning protection system;
- i. Fire monitoring and suppression systems;
- j. Pole mounted and building mounted exterior lighting; and
- k. Roads and parking lots.

The computational environment consists of the installed Customer IT, networking and telecommunications systems and the NCCIPS critical facility infrastructure. See TE-2, IT Matrix, for details. The major categories of resources are:

- l. Infrastructure networking (fiber optic cable plant and patching support);
- m. Share Point management tools for workflow, acquisition and operations management;
- n. NCCIPS TechDoc for documentation management and other support applications;
- o. Voice Over Internet Protocol (VOIP), fire, security (camera and badging) and other equipment, which supports the operation of the data center; and
- p. Data Center Integrated Management tools.

2.0 PROGRAM MANAGEMENT

The Contractor shall perform NCCIPS program management functions including, but not limited to: resource management, scheduling, subcontract management, quality control, business management, risk management, facilities support, safety and occupational health management, records management, security support, workload data and metrics reporting, business continuity planning and execution, acquisition support, logistics support (property, shipping, receiving, mail distribution), support to stakeholder/employee communications, and new business development. The Contractor shall ensure that its subcontractors perform these same program management functions, as applicable to their specific responsibilities. The Contractor shall establish a management approach that embodies sound financial management concepts and controls, stewardship, flexibility, strong customer-focus, and best business practices. The Contractor shall perform the activities/functions outlined in this PWS in accordance with all applicable Federal, state, and local requirements.

Program Management shall include supporting a Government Point of Contact (POC) and supporting a Customer Advisory Forum for the purpose of communicating NCCIPS announcements common to all Customer tenants (budgetary, maintenance related, security related, and all other matters related to the support, operations, maintenance and management of the Data Center Facility), soliciting new and revised Customer agency service level requirements, and communicating Customer global requirements (common security controls, sensitive information handling, and site visit request procedures).

In support of the NCCIPS mission, the Contractor shall develop and maintain a leadership culture characterized by process-centered, productivity-based, and customer-focused values that are understood, acted upon, and routinely measured at all levels of the organization. Key elements of strategic alignment are:

- a. NCCIPS mission and strategic objectives;
- b. Business values that align with and support the mission and strategic objectives;
- c. Measurement methods to baseline and periodically measure strategic objective alignment; and
- d. A well-developed system that mobilizes leaders at every level to foster the mission and the strategic objectives.

Areas of special interest requirements shall include:

- e. Forecast future service infrastructure capacity requirements;
- f. Take initiative in framing issues, providing information, and making recommendations;
- g. Recommend enterprise strategies to ensure adequate links between current and future business and technology needs;
- h. Stay abreast of changing enterprise business needs and technology capabilities; and
- i. Make recommendations for changes in a strategic direction as appropriate to leverage technology effectively in service delivery.

2.1 CONTRACTOR EMPLOYEE REQUIREMENTS

2.1.1 Contractor Personnel Training

NASA will periodically provide mandatory training for all NCCIPS employees in areas such as safety, security, occupational health, IT security, and fire prevention. When mandatory training sessions are held, the Contractor shall ensure that its employees attend and shall maintain attendance records.

Training will include, but may not be limited to:

- a. Security orientation training that covers internal security requirements and procedures outlined in NASA Procedural Requirements (NPR) 1600.1 “NASA Security Program Procedural Requirements,” NPR 1600.2 “NASA Classified National Security Information (CNSI),” and NPR 1600.3 “Personnel Security,” as amended;
- b. Safety orientation training that covers occupational health and safety requirements outlined in NPR 8715.3 “NASA General Safety Program Requirements,” as amended;
- c. IT Security Training, in accordance with NPR 2810.1 “Security of Information Technology,” as amended;
- d. NSSC Plan (NSPLN) 1040-0001 “NSSC Emergency Action Plan,” which includes:
 - i. Fire Emergencies,
 - ii. Emergency Evacuation,
 - iii. Severe Weather,
 - iv. Bomb Threat,
 - v. Work Place Violence, and
- e. Federal and NASA laws/policies regarding Data Center management; and
- f. NASA Emergency Notification and Accountability System.

NASA may provide other on-site training that the Contractor may be invited to participate in on a space available basis; however, the Contractor is responsible for ensuring all its employees are fully trained and qualified for the positions they fill without any reliance on Government provided training. System training modules, courses in the Agency’s Learning Management System (LMS), and other existing training materials will be available for Contractor use. The Contractor is also responsible for developing training when deemed necessary, including video production, PowerPoint, written, and instructor led training as appropriate. The Contractor shall provide support for all Contractor and Civil Servant (CS) personnel participating in NASA sponsored training courses. All NASA provided Contractor training shall be recorded in the Agency’s LMS and shall be accessible by the Government (Data Requirements Description (DRD) 2.1-1).

2.1.2 Contractor Employee Data

The Contractor shall maintain an on-line organization chart and employee roster, accessible to the Contracting Officer’s Representative (COR) and other designees, of individuals who perform work under this PWS (DRD 2.1-2). The roster shall be updated for each change within five (5) business days of its occurrence. The Contractor shall include the following information in the roster:

- a. Employee Name;
- b. Labor Category/Job Title;
- c. Organizational Entity;
- d. Phone/Cell Number;
- e. Facsimile Number;

- f. Site Location/Address;
- g. Room Number;
- h. Email address; and
- i. Emergency Point of Contact.

2.1.3 Contractor Personnel Access

NASA reserves the right to control and deny access to its facilities, property, and data, including those that are identified in this PWS. Access privileges are tailored to individual NCCIPS personnel responsibilities. NASA will have the final authority in determining access privileges. NASA's exercise of its right to grant and revoke access for particular individual(s) to its facilities, property, or information will not constitute a breach or change to the contract, regardless of whether said individual(s) is/are employed by the Contractor, and regardless of whether said individual(s) is/are precluded from performing work under the PWS.

2.1.4 Designation of Position Risk Level

Prior to employment, all NCCIPS Contractor staff shall have an appropriate personal background investigation (security screening) and receive a favorable adjudication determination from the designated servicing NASA Center Office of Protective Services (OPS.) The Contractor shall comply fully with NASA Federal Acquisition Regulation (FAR) Supplement 1852.204 and NPR 1600.3 "Personnel Security," as amended. At a minimum, the Contractor shall:

- a. Consult with the SSC OPS and the NCCIPS CS Security Coordinator to determine the designated risk and sensitivity levels of each Contractor position established as part of the contract per the NASA FAR Supplement and NPR 1600.3 "Personnel Security," as amended (DRD 2.1-3);
- b. Conduct appropriate background investigations based on the individual's position risk level as outlined in NPR 1600.3 "Personnel Security," as amended; and
- c. Upon receipt, forward the results of completed background investigations to the designated servicing NASA Center OPS for final access determination.

2.1.5 Equal Employment Opportunity

The Contractor shall comply with all Equal Employment Opportunity (EEO) laws and regulations and integrate the principles of Diversity and Inclusion into the Contractor way of doing business. The Contractor shall create and maintain an environment in which each individual is valued, respected and enabled to reach his/her highest potential. The Contractor shall ensure all employees are provided opportunities to contribute towards fulfilling the NCCIPS mission. The Contractor shall submit EEO and Diversity and Inclusion information in accordance with DRD 2.1-4.

2.2 RECORDS, DOCUMENTS, AND FORMS MANAGEMENT

The Contractor shall provide comprehensive management of the records, documents and forms associated with NCCIPS operations in accordance with Federal requirements, NASA policy, and NSSC policy. The Contractor shall develop a plan for documentation development, organization and management (DRD 2.2-1). The plan shall address and assure timely and accurate organization, identification, marking, management, preservation and disposition of NASA documentation, records and forms regardless of format in accordance with NPR 1441.1 "NASA Records Management Program Requirements" and NSSC Process Work Instruction (NSPWI) 1420-0001

“NSSC Forms Management Process Work Instruction,” as amended. To support these requirements the Contractor will:

- a. Maintain a complete Master Records Index for all NCCIPS records, including photographs and video used to document NCCIPS;
- b. Maintain an index of NCCIPS forms;
- c. Provide detailed organization, numbering, and structure of documents within the NCCIPS Electronic Library to ensure simplicity of document search and retrieval and meet records retention requirements per NASA Records Retention Schedules (NRRS) 1441.1 “NASA Records Retention Schedules,” as amended;
- d. Ensure that electronic systems meet records retention requirements with required metadata to identify records;
- e. Conduct internal reviews, support audits of records, and provide detailed reports (DRD 2.2-1) with corrective action plans to the NSSC Records Manager; and
- f. Process records for transfer to Government records storage facilities and for disposal.

The Contractor shall utilize the NSSC Forms Program ensuring access to electronic forms for NCCIPS personnel. The Contractor shall ensure the records management plan coincides with the NSSC document management plan and that records are properly incorporated into the NCCIPS TechDoc document management system as applicable.

2.3 REPORTING / INFORMATION REQUESTS

The Contractor shall respond to all data calls and requests for reports and information, including scheduled and ad hoc requests, from the COR or NSSC Contracting Officer (CO). Requested reports may contain Sensitive But Unclassified or Controlled Unclassified Information as defined in NASA Interim Directive (NID) 1600.55 “Sensitive But Unclassified Information,” NID 1600.54 “Safeguarding Sensitive But Unclassified Information,” and NASA Requirement Waiver (NRW) 1400-48 “Requirement Waiver Request for NID 1600.54 and NID 1600.55,” as amended.

The Contractor shall notify the COR of requests received from other sources prior to responding to those requests. The Contractor shall provide support to the NSSC CO and the NSSC Freedom of Information Act (FOIA) Officer by gathering documents for FOIA requests. The Contractor shall refer any FOIA inquiries that it receives to the NSSC CO and the NSSC FOIA Officer.

2.4 NCCIPS SITE VISITS / INSPECTIONS

The Contractor shall support all NCCIPS site visits and inspections when notified by the COR. The Contractor shall cooperate with visiting Government and Contractor personnel conducting official inspection visits, audits and surveys. The Contractor shall notify the COR of planned visits, investigations, or corrective actions required by Federal, state, and local agencies. The Contractor shall notify the COR of the unannounced arrival of representatives from any external Federal, state, or local government agency or organization. The Contractor shall provide support for coordinating site visits, updating informational display monitors and bulletin boards, and scheduling conference rooms and briefings.

2.5 PERFORMANCE METRICS REPORTING

The Contractor shall provide and maintain a system for management and reporting of performance metrics. All performance management and reporting shall be in accordance with Government-defined performance standards identified in Attachment J-3, Performance Requirements Summary, and in Attachment J-2, Data Requirements List. The Contractor shall prepare and submit a Performance Measurement Plan reflecting the approach for capturing and reporting all performance measures (DRD 2.5-1). The Contractor shall capture and report all performance measures as described in DRD 2.5-2 and the reports shall be available via the NCCIPS TechDoc system.

Attachment J-3 provides a listing of minimum Service Level Indicators and performance measures that the Government intends to use to verify and evaluate Contractor performance. The Contractor shall develop, maintain, analyze, and report performance metrics for work described in this PWS. The Contractor shall report these metrics as part of management reviews. The Contractor shall provide objective and measurable metrics tracking and reporting. Based on its analyses and the Government's analyses of metric performance, the Contractor shall report existing or potential problem areas with recommended solutions.

The Contractor shall provide a monthly PWS-based status report and briefing to the Government. The Contractor shall conduct Weekly Status Reviews (WSR) for the Government that provide performance status reports/updates in the key areas of program management and facility management. The format of the WSRs will be determined by the Contractor and the Government during contract phase in. The WSR is not expected to last more than one hour per week. More frequent, specific ad hoc operational briefings may be required such as when new Customers transition to the NCCIPS.

2.6 INNOVATION AND CONTINUOUS PROCESS IMPROVEMENTS

The Contractor shall continuously assess and benchmark internal processes to look for ways to improve facility and infrastructure services in accordance with DRD 2.6-1. The Contractor shall continuously improve business processes and increase productivity to optimize the quality and efficiency of services provided to the NCCIPS, its Customers, and stakeholders. Continuous improvement projects shall be initiated by the Contractor to increase productivity, reduce costs, or increase quality. Where such projects impact NCCIPS Customers or CS staff, Government approval based on a cost benefit analysis shall be required. NCCIPS CS staff may also initiate improvement projects that may require Contractor support. Examples of such projects may be improved Power Usage Effectiveness factor, reduced costs in physical security, and/or consolidation with NSSC systems/processes as applicable.

2.7 SECURITY

The Contractor shall identify a primary and alternate NCCIPS Contractor Security POC who shall be responsible for interfacing with the CS NCCIPS Security Coordinator. The Contractor shall provide this information to the COR within ten (10) calendar days of award of contract. In the event that the POC must be changed, the Contractor shall notify the COR of the proposed change no later than fifteen (15) calendar days prior to proceeding with the change. The NCCIPS Contractor Security POC shall be responsible for interfacing with the appointed NCCIPS Security

Coordinator and the SSC Office of Protective Services (OPS) in regards to the coordination of NCCIPS security requirements associated with facility physical protection, personnel credentialing, and access control. The NCCIPS Contractor Security POC shall provide all applicable information and documentation to the NCCIPS Security Coordinator and SSC OPS necessary to support facility physical protection, personnel credentialing, and access control. The NCCIPS security program shall be in compliance with NPR 1600.1 "NASA Security Program Procedural Requirements," NPR 1600.2 "NASA Classified National Security Information (CNSI)," NPR 1600.3 "Personnel Security," and NPR 3730.1 "NASA Suitability Program," as amended. Determinations as to the effectiveness of the NCCIPS security program will be conducted by the NASA OPS and/or the SSC OPS. The Contractor shall comply with the Intelligence Community Standards, specifically Intelligence Community Directive 705 as well as Executive Order 12333 "United States Intelligence Activities," and Executive Order 13526 "Classified National Security Information," as amended, which establish standards for security clearances. The Contractor shall also be responsible for adhering to the security requirements of the NCCIPS Customers in the event that Customer security requirements are more stringent than required by NASA.

The Contractor Security POC will provide all Contractor onboarding documentation for new Contractor personnel as required by NPR 1600.1 and the SSC OPS.

The Contractor shall submit a comprehensive NCCIPS IT Security Management Plan, to include key control, media access and protection (DRD 2.7-1).

2.7.1 Physical Security

Physical security requirements necessary to ensure appropriate protection for NCCIPS personnel, NCCIPS Customers, facilities, and information will be established and managed by the Contractor Security POC in accordance with NPR 1620.3 "Physical Security Requirements for NASA Facilities and Property," and NPR 2810.1 "Security of Information Technology," as amended.

2.7.2 Personnel Security Clearances

In accordance with position risk and sensitivity, all NCCIPS employees, at a minimum, must be adjudicated and subsequently obtain a SECRET clearance. In coordination with the NCCIPS Security Coordinator, Contractor personnel whose duties require access to CNSI shall submit requests for appropriate security clearances in accordance with the procedural requirements established in NPR 1600.3 Chapter 3 "Personal Security Investigations for National Security Positions," and NPR 1600.2 "NASA Classified National Security Information (CNSI)," as amended. The NASA OPS has final authority when determining an individual's access to CNSI.

2.7.3 Key Control, Media Access and Protection

The Contractor shall identify a primary and alternate Key Control POC. The Contractor shall provide this information to the COR within ten (10) calendar days of award of contract. In the event that the POC must change, the Contractor shall notify the COR of the proposed change no later than fifteen (15) calendar days prior to proceeding with the change. The Contractor Key Control POC shall be responsible for managing, controlling and issuing keys to NCCIPS personnel. This includes interfacing with SSC locksmith personnel for re-keying or physical core replacement if required.

The Contractor shall establish and maintain procedures in accordance with NPR 1620.3 “Physical Security Requirements for NASA Facilities and Property,” as amended, to ensure that any NCCIPS access media, such as keys, badges, and pass codes issued to Contractor employees are not lost, misplaced, or used by unauthorized persons. Contractor procedures shall include the requirement to maintain current, accurate, comprehensive records that include the following data:

- a. The names of individuals to whom access media are assigned; and
- b. Access media control numbers.

Contractor access control records shall be electronically accessible to the COR. The Contractor shall not duplicate NASA-issued access media without written authorization from the NCCIPS Security Coordinator. Use of NASA-issued access media is restricted to the Contractor employees to whom they are issued. The Contractor shall not permit entrance to controlled or locked areas by any person other than Contractor personnel engaged in the performance of NCCIPS work.

The Contractor shall report any occurrence of lost media or unauthorized access to the NCCIPS Security Coordinator immediately upon discovery of the loss or unauthorized access. Additionally, for each lost access media occurrence, the Contractor shall forward a written report to the NCCIPS Security Coordinator by close of business the next business day, unless directed to do otherwise by the COR. Lost access media reports shall contain the access media control number, location(s) accessed by the person who lost the access media, date the access media was discovered missing, name of person who signed for the access media, immediate risks and mitigation, and any other relevant details (DRD 2.7-2).

The Contractor shall establish and maintain procedures to ensure that NCCIPS storage media, issued to or utilized by Contractor employees is secured per NPR 2810.1 “Security of Information Technology,” as amended. The Government will provide information to the Contractor regarding items issued to NCCIPS employees under other contracts.

2.7.4 On-Boarding, Departures and Inter-Office Transfers

The Contractor shall develop processes and associated instructions for coordinating with NSSC IT Security and the IT Asset Manager for all on-boarding, departures, inter-office transfers, and transfers to other contract affiliations at SSC. All badged access to the NCCIPS campus and all NASA Access Management System (NAMS) access rights shall be removed for departing Contractor personnel. Applicable NAMS access and approval controls (NAMS workflow roles / provisioning) will be modified as needed for inter-office transfers and transfers to other contract affiliations at SSC.

2.8 SAFETY AND OCCUPATIONAL HEALTH

The Contractor shall identify a primary and alternate Contractor Safety POC. The Contractor shall provide this information to the COR within ten (10) calendar days of award of contract. In the event that the POC must change, the Contractor shall notify the COR of the proposed change no later than fifteen (15) calendar days prior to proceeding with the change.

The Contractor Safety POC shall be responsible for interfacing with the appointed NASA CS NCCIPS Safety Coordinator and the SSC Office of Safety and Mission Assurance (OSMA) in regards to safety and occupational health issues that may affect NCCIPS personnel and operations. The Contractor shall conduct all work in a safe manner and in compliance with, at a minimum, Occupational Safety and Health Administration (OSHA) Standards, Environmental Protection Agency requirements, and NPR 8715.3 “NASA General Safety Program Requirements,” as amended. If a safety requirement cannot be met, the Contractor shall submit a “deviation/waiver” request for approval to the NCCIPS Safety Coordinator. The Contractor shall maintain and report to NASA, in accordance with OSHA and NASA documentation requirements, an accurate record of work-related illnesses, accidents, and incidents causing injury or death and accidents and incidents resulting in damage to Government property, supplies, and equipment. All mishap and close call reporting and handling shall be conducted in compliance with NPR 8621.1 “NASA Procedural Requirements for Mishap and Close Call Reporting, Investigating, and Recordkeeping,” as amended. The Contractor shall immediately report to NASA all “Unsafe” or “Unhealthy” conditions within the workplace. The Contractor shall submit an incident report for each incident resulting in an accident, near miss, or personal injury, in accordance with DRD 2.8-1.

The Contractor shall establish and maintain an employee safety and health committee that will formally meet at least once per month. The Contractor Safety Committee will be chaired by the Contractor Safety POC and minutes of the meetings shall be recorded in accordance with NASA recordkeeping guidelines and shall be considered official records. The primary objective of the Contractor Safety Committee is to enhance employee safety and health. The personal safety and health of each Contractor employee is of primary importance and the prevention of occupationally-induced injuries and illnesses is the Contractor Safety Committee’s principal purpose. To the greatest degree possible, the Contractor Safety Committee shall provide Contractor personnel a voice to ensure all mechanical and physical facilities required for personal safety and health are controlled and maintained in keeping with the highest standards.

In carrying out its primary mission, the Contractor Safety Committee shall be committed to five basic principles:

- a. Help to formulate and disseminate policies, practices and procedures that promote Safety and Health at the NCCIPS;
- b. Consult with the NCCIPS Safety Coordinator on questions or issues concerning safety and health policy, practice, or procedure;
- c. Assist the CS and Contractor management in the planning of corrective actions concerning identified occupational health and safety issues;
- d. Act as a problem solving group to help with the identification and control of hazards; and
- e. Conduct and document monthly facility safety and health inspections and facilitate actions to resolve those hazards identified as health and safety discrepancies.

For fire safety, the Contractor shall comply with all applicable Federal, State, and local requirements pertaining to Fire Protection and Life Safety, and NASA Standard 8719.11 “NASA Safety Standard for Fire Protection.”

For electrical safety, the Contractor shall comply with National Fire Protection Association (NFPA) Standard 70E “Standard for Electrical Safety in the Workplace.”

2.8.1 Disaster Recovery and Emergency Situations

Emergency situations (including, but not limited to, disasters of any kind whether natural disasters, accidents, or terrorist-related in nature) may necessitate the Contractor to operate on an extended or curtailed schedule (including days or shifts not normally scheduled) at a different level of service. This includes changes in security condition levels for the NCCIPS facilities, which may impact normal operations.

Extreme weather and other hazardous conditions may warrant temporary office evacuation or NCCIPS closure with the exception of essential personnel. The Contractor shall respond to extreme weather and other hazardous conditions according to COR direction, and shall inform all Contractor employees of these instructions.

Announcements of facility closures will be made in the following manner:

- a. During Normal Duty Hours, notification will be given through normal chain of management ; and
- b. During Non-Duty Hours, notification will be provided through the NASA Emergency Notification System.

Subsequent transmissions of pertinent information or updates concerning facility or Center closures will continue using the alert and notification procedures found within Section 2.12 (Communication to NSSC Employees) of NSBCP-1040-0001 “NSSC Business Continuity Plan (BCP).”

Facility closings shall not interfere with the Contractor operation and/or maintenance of the critical systems. Such interruptions will be considered when assessing Contractor performance for the affected period. The Contractor shall provide business strategies, resources and procedures required for the immediate response and subsequent recovery from short or long term business interruptions, such as: loss of critical IT services, loss of building access or a physical facility disruption in accordance with Business Continuity and Disaster Recovery Planning as described later in this section of PWS 2.8.1. Furthermore, all Contractor employees identified as essential personnel shall remain on duty or report for duty in accordance with the procedures outlined in the NSSC BCP.

The NCCIPS campus must be maintained 24x7x365. In the event of a Center closing, the Contractor shall identify the specific personnel needed to perform the technical functions of a normal shift. Typically this consists of an operations supervisor, operators, electricians, safety manager, generator technician, general maintenance technician, and IT technician. The Contractor Program Manager will identify by name, the essential personnel needed to maintain operations for each event of a Center closure. All essential personnel are to be reported as “excepted personnel” for Center closures in accordance with SSC procedures.

Unless directed otherwise by the COR or CO, the Contractor shall participate in all scheduled and unscheduled fire drills or other scheduled safety and emergency training exercises. This effort includes the planning, scheduling, and execution of fire drills and emergency training exercises which may necessitate diversion of personnel from normal duties, during the participation in these

drills. However, this effort shall be coordinated to prevent interruption of key services or interference with key NCCIPS operations.

The Contractor shall support all emergency preparedness activities to include the review, planning, and execution of NCCIPS consolidated emergency preparedness plans as noted later in this section of PWS 2.8.1.

The Contractor shall support the NSSC and NCCIPS management team in disaster recovery activities pertaining to the NCCIPS and Customer systems. The Contractor shall, along with the CS, jointly develop an IT Disaster Recovery Plan in accordance with DRD 2.8-2. The Contractor shall conduct and participate in test exercises as required by applicable Disaster Recovery Plans.

The Contractor shall provide project management support in developing and maintaining NCCIPS strategic plans, integrated project schedules, and consolidated emergency preparedness plans, as required. The NSSC Consolidated Emergency Preparedness Plans include the following:

- c. NSBCP-1040-001 “Business Continuity Plan” – Document designed to provide a general overview of the business strategies, resources and procedures required for the immediate response and subsequent recovery from short or long term business interruptions, such as:
 - i. Loss of critical IT services; and
 - ii. Loss of building access or a physical facility disruption.
- d. NSPLN-1040-001 “Emergency Action Plan” – Documented procedure(s) to provide affected employees with a planned response to localized emergencies in order to protect lives and property. Emergencies can arise from natural disasters or from human activities that are intentional or unintentional and have the potential for creating devastating destruction.
- e. IT Disaster Recovery Plan in accordance with DRD 2.8-2 – Documented process or set of procedures to recover and protect a business IT infrastructure in the event of a disaster.

2.8.2 Occupational Health

The Contractor shall conduct all work in a safe and healthy manner in compliance with, at a minimum, OSHA, Department of Labor regulations and NPR 1800.1 “NASA Occupational Health Program Procedures,” as amended. The designated Contractor Safety POC will initiate, coordinate and document all Americans with Disabilities Act ergonomic assessment requests from Contractor personnel.

2.9 RISK MANAGEMENT AND MITIGATION

The Contractor shall perform and maintain a comprehensive risk management and mitigation program for abating all identified management and technical risks, including those for new or revised projects and systems; disaster preparedness; labor relations; loss of skill sets and institutional knowledge to support functional activities; information systems infrastructure; disruption to current service levels; and the organizational change necessary to implement NSSC’s vision for providing outstanding customer service. The Contractor shall provide a Risk Management Implementation Plan in accordance with NPR 8000.4 “Agency Risk Management Procedural Requirements” and with NSPLN-1240-0001 “NSSC Risk Management Plan,” as amended (DRD 2.9-1).

2.10 QUALITY CONTROL

The Contractor shall establish and conduct a Quality Management System for all services provided under this contract that complies with the guidance in International Organization for Standardization (ISO) 9001:2015 - Quality Management (ISO certification is not required). The Contractor shall develop a Quality Control Plan that complies with ISO standards. The Contractor's Quality Control Plan shall be documented in accordance with DRD 2.10-1.

The Contractor shall provide support for all aspects of quality control including building drawings, photographs and video as needed to document as-is and as-built NCCIPS configurations, written documentation, and deliverables.

2.11 CONTRACTOR FINANCIAL MANAGEMENT

PWS 2.11 FINANCIAL MANAGEMENT

The Contractor shall submit detailed and accurate invoices in accordance with the clauses contained in this contract. The invoice (or its supporting documentation) shall be separated in accordance with the following NCCIPS Work Breakdown Structure (WBS):

WBS DESCRIPTION

- 10 Operations and Maintenance Summary
 - 11 Core Services **(Base Year)**
 - 12 Facilities & Equipment Maintenance **(Base Year)**
 - 13 Special Projects **(Base Year)**
 - 14 General Materials & Supplies **(Base Year)**
- 2.0 Electrical Support Contracts – includes maintenance contracts for HVAC, Generators, and Power Distribution Systems. **(Base Year)**
- 30 Customer Demand - a separate WBS shall be established for each customer
 - 31 Demand Acquisitions - each customer shall have one sub-level WBS for Acquisition Support procurements. **(Base Year)**
 - 32 Demand Labor - each customer shall have one sub-level WBS for demand (non-project) labor orders. **(Base Year)**
 - 33 Demand Projects - each customer will have one to many sub-level WBS for demand projects (each reimbursable customer project shall be invoiced with a specific project WBS and all project costs shall be charged to the project WBS). **(Base Year)**

The following table provides an example customer WBS breakdown for DHS.

WBS	Description
.03.03	DHS Orders
.03.03.01	DHS Demand Acquisitions
.03.03.02	DHS Demand Labor
.03.03.03	DHS Demand Projects
.03.03.03.01	DHS Project 181001R
.03.03.03.02	DHS Project 190102R
.03.03.03.03	DHS Project 190601R
.03.03.03.04	DHS Project 190603
.03.03.03.05	DHS Project 190701
.03.03.03.06	DHS Project 190801
.03.03.03.07	DHS Project 190802
.03.03.03.08	DHS Project 190805
.03.03.03.09	DHS Future Project
.03.03.03.10	DHS Future Project
.03.03.03.11	DHS Future Project
.03.03.03.12	DHS Future Project
.03.03.03.13	DHS Future Project
.03.03.03.14	DHS Future Project
.03.03.03.15	DHS Future Project

The Contractor deliverables shall include Microsoft Excel versions for integration into the NSSC systems. Immediately following contract award, the Contractor shall work with the NSSC Budget & Accounting Division to develop and agree acceptable formats.

2.12 ACQUISITION SUPPORT

The Contractor shall support DEMAND activities and the provisioning of General Materials & Supplies per PWS Section 3.1.1.3 on a reimbursable basis as directed by the COR. Monthly reports in accordance with DRD 2.12-1 shall be provided reflecting the status of procurements in the acquisition process through completion of activities. The actual monthly acquisition costs in DRD 2.12-1 must reconcile to the invoiced acquisition costs for each reporting period. All acquisition actions shall be conducted in accordance with applicable FAR sections. The parties expressly agree that the offered and accepted Firm Fixed Price (FFP) for CORE SERVICES includes all costs incurred or paid by the Contractor, including but not limited to material handling charges, overhead, general and administrative costs, or profit, that are in any way associated with the Contractor's purchase or provision of supplies and services related to DEMAND activities.

The Government will reimburse the Contractor only for the actual price paid for those supplies and services that are expressly identified as a DEMAND activity by this contract up to the level previously approved by the CO. "Actual price" paid by the Contractor for such supplies and services includes tax paid, if any, and reduced by any and all credits and rebates, whether accrued or realized, associated with the supplies and services provided. "Actual price" does not include

material handling charges, overhead, general and administrative costs, profit, or any other indirect cost that is in any way associated with the Contractor's purchase or provision of such supplies and services.

To be eligible to receive reimbursement for equipment, services and supplies related to DEMAND activities and obtained in support of this contract, the Contractor must obtain at least three quotes for each transaction in excess of \$10,000 to ensure that adequate price competition was sought or the Contractor must provide an acceptable justification as to why it was impracticable to do so. The Contractor shall provide this documentation to the CO for approval in accordance with FAR 52.244-2 (Subcontracts), or as directed by the CO, prior to obtaining these services and supplies. The Contractor shall maintain documentation of all reimbursable purchases until three years after the contract is completed and shall provide access to and copies of such documentation when requested by the CO.

The CO may reduce the reimbursement by any amount above that which the CO finds, in his/her sole discretion, is greater than that which is fair and reasonable for the supplies or services provided, giving due consideration to the facts and circumstances prevailing at the time that the Contractor procured the supplies and services. Disputes as to the amount by which any reimbursement is reduced shall be resolved in accordance with the "Disputes" clause of the contract. It shall be the Contractor's burden to demonstrate that the price it paid for reimbursable supplies and services was fair and reasonable.

When the Contractor expects total funding expended for either the Customer Orders CLINs or the Special Projects CLINs to reach 75 percent of the total funds available, the Contractor shall notify the CO and the COR and any other Government official identified by the CO. The notice shall state the estimated amount of additional funds required to continue performance for the period specified in the Schedule. The Contractor shall not exceed or incur costs that exceed the amount of funding stated on each CLIN at the time a reimbursable item is ordered.

The Government is not obligated to reimburse the Contractor in excess of the funded amount stated in the Schedule under the Customer Orders CLINs and the Special Projects CLINs.

The Contractor is not obligated to continue performance of any work under this Contract or otherwise incur costs in excess of the funded amount stated in the Schedule under either the Customer Orders CLINs or the Special Projects CLINs unless the CO notifies the Contractor that the funded amount stated in the Schedule under the applicable CLIN has been increased. Notification shall be in writing. In the event notification is made orally, such notification shall be followed up in writing within two working days.

No notice, communication, or representation from any person other than the CO shall affect the Government's obligation to reimburse the Contractor.

Change orders shall not be considered an authorization to exceed the funded amount stated in the Schedule under the Customer Orders CLINs and the Special Projects CLINs unless they contain a statement expressly increasing the funded amount in the subject CLIN by a sufficient amount to cover the change order.

2.13 PHASE-IN PLAN

The Contractor shall perform the requirements of this PWS in accordance with the Contractor Phase-in Plan. The phase-in shall be as seamless as possible with no disruption to services. The Contractor shall submit a phase-in plan with its technical proposal and the approved phase-in plan will be incorporated into the contract as Attachment J-16, Phase-In Plan.

2.14 NCCIPS CUSTOMER SUPPORT HOURS

The Contractor shall provide the capability to support NCCIPS Customers 24x7x365. Some activities require other than normal business hour processing. IT and facility upgrades and maintenance shall be performed during times that cause minimal disruption to NCCIPS and Customer operations. As such requirements are identified, the Contractor shall promptly notify the COR or CO. Normal Business Hours are defined as Monday – Friday, 8a – 5p Central.

2.15 CERTIFICATIONS, LICENSES, AND PERMITS

The Contractor shall obtain all necessary certifications, licenses, and permits required for performance of work and for complying with all applicable Federal, state, and local laws and regulations.

2.16 ENVIRONMENTAL REQUIREMENTS

The Contractor shall comply with all Federal, State, and local environmental protection laws, regulations, and standards. The Contractor shall limit the quantity of Hazardous Materials on-site to the minimal amount necessary for normal business operations.

2.17 EMERGENCY CONTRACTING

The Contractor's obligation may include resolution of unusual or emergency situations. The Contractor may be required to assist NASA, within the general scope of work, but in currently unidentified ways, in preparation for, or in response to emergencies. Obligations under this requirement shall only arise when one or more of the criteria at FAR Part 18.001, enabling NASA to utilize "Emergency Acquisition Flexibilities," are met. If the emergency preparedness and response requirements result in changes to the contract, all contract adjustments will be processed in accordance with the Changes clause of this contract.

2.18 PROPERTY MANAGEMENT

The Contractor shall provide comprehensive property management of NCCIPS Government Property and IT assets in accordance with Federal Acquisition Regulations, NASA Property Policy Directives, NASA Procedural Requirements, and contractual requirements. Overall Property Custodian responsibilities will be designated to a Contractor employee. The Property Custodian shall be trained in the use of NASA's Plant Property and Equipment (PP&E) system and shall ensure users of property are properly identified in the system. The Property Custodian has all responsibilities as identified by NASA policy to ensure Government Property is properly marked and its locations tracked. Local PP&E system management is provided by NASA and the Property Custodian must work closely with the NASA Logistics Management personnel. The Contractor shall provide a Property Management Plan (PMP) (DRD 2.18-1) in accordance with FAR Part 45. The Contractor's performance of the PMP will be audited in accordance with NASA Policy

Directive 4500.1 “Administration of Property in the Custody of Contractors.” The Contractor shall provide a separate IT Seat and System Hardware Asset Plan (DRD 2.18-2) to track locations and users of Seat assets in the NCCIPS occupied facilities. IT Seats refer to the desktops, laptops, smart phones and other standard IT equipment obtained from the Agency’s Information Technology Services Program.

The Contractor shall provide property management in regards to inventory and inventory ownership documentation in support of the NCCIPS and the SSC telecommunications building to include the following:

- a. Ensure that acquisitions are contractually authorized and consistent with its engineering, production planning, and material control operations;
- b. Ensure controlled receipt for all property. Establish a process for identifying items received for purpose of establishing control of Government owned equipment. In addition, a process shall be established for recording and controlling consumable materials received;
- c. Establish a property management system to maintain complete, current, and auditable records for property accountable under the Contractor stewardship;
- d. Maintain an inventory of all designated equipment identifying equipment as NCCIPS or Customer owned;
- e. Efficiently manage temporary and long term storage of both NCCIPS and Customer property in the NCCIPS receiving facility, storage bunkers, and incidental storage areas;
- f. Provide inventory reports in accordance with DRD 2.18-3 (reference NPR 4100.1 “NASA Supply Support and Material Management” and NPR 4200.1 “NASA Equipment Management Procedural Requirements”);
- g. Dispose of communications-computer systems no longer needed or serviceable in accordance with Government requirements; reference NPR 4300.1 “NASA Personal Property Disposal Procedural Requirements;”
- h. Schedule and perform property control audits as defined in the PMP and provide results to the Government (DRD 2.18-4);
- i. Assist NASA in documentation and transfer of property from NCCIPS Customers. Identify, disposition, and retain or dispose of property as appropriate;
- j. Safely transfer/move NCCIPS and Customer property as required throughout the NCCIPS campus and the telecommunications building;
- k. Safely and effectively utilize and manage use of General Services Administration provided vehicles assigned to NCCIPS (e.g., pickup trucks and automobiles) for performing NCCIPS-related functions;
- l. Maintain vehicle usage logs in accordance with DRD 2.18-5;
- m. Travel to local area vendor locations to pick up property and supplies purchased for NCCIPS use; and
- n. Ensure applicable NCCIPS Work Instructions (WI) and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List.

2.19 GENERAL MAIL SERVICES

The Contractor shall be responsible for all NCCIPS mail sorting, mail distribution and package shipping and receiving. Incoming and outgoing mail and packages shall be delivered/picked up

daily, Monday through Friday. This is exclusive of equipment and parts shipping and receiving requirements described in PWS Section 3.1.1.1.

2.20 NEW BUSINESS SUPPORT

The Contractor shall provide support to the NSSC's efforts to explore new business opportunities and attract new NCCIPS Customers. Support includes:

- a. Developing business cases, cost and time estimates to support new opportunities;
- b. Establishing transition strategies that address processes, metrics, timelines and information technology requirements; and
- c. Implementing innovative service delivery approaches that limit Government investment in application and infrastructure costs.

2.21 PRINT AND COPY

The Contractor shall comply with the printing, duplicating, scanning, and copying/printing requirements in NPD 1490.1 "NASA Printing, Duplicating, and Copying Management," as amended.

2.22 WASTE MANAGEMENT

The Contractor shall comply with the waste prevention and recycling guidelines in Executive Order 13101 "Greening the Government through Waste Prevention, Recycling, and Federal Acquisition."

2.23 RIGHTS IN DATA

The architecture, implementation plan, unique capabilities, materials and other items designed and/or developed for this PWS and related projects and the complete data rights to these items shall be the sole property of the U.S. Government. All information derived and used for development is considered Government proprietary and shall not be used or shared and is considered intellectual property of the U.S. Government and shall not be released. After completion and acceptance, all information held by the Contractor shall be deleted from the Contractor's system(s) and the Contractor Security POC will validate in writing to the Government COR that all documentation and information derived from the contract has been purged from all systems, whether administrative or program related. Only the contractual and budgetary (cost, invoicing, etc.) information required by law shall be retained by the Contractor.

2.24 PLACE OF PERFORMANCE

Work shall be performed on-site at the NCCIPS campus at Stennis Space Center, MS, and/or at other designated locations within the continental United States. NCCIPS may define other specific facilities within the continental United States, dependent on need, during performance of the requirements. The Contractor is notified that some Contractor employees may be working in off-site government or commercial facilities/areas with varying safety standards and conditions, such as where hazardous materials are stored or in use; construction zones; areas where heavy machinery and equipment are in use, etc. and employees must comply with safety codes, regulations and standards appropriate for the conditions.

2.25 KEY PERSONNEL

The Government desires that Key Personnel be assigned for the duration of the PWS. The following positions are considered key personnel with minimum qualifications:

- a. Program Manager: The Program Manager shall be designated by the Contractor and shall have complete authority to act for the Contractor in every detail during the period of performance of this PWS. The Program Manager's physical location shall be at the NCCIPS facility. The Program Manager shall possess at a minimum five years of recent experience in the management and supervision of data center facility operations and maintenance for Tier III, multi-tenant, government data centers or equivalent experience. The Program Manager shall be able to:
 - i. Analyze new and complex project related problems;
 - ii. Create innovative solutions involving large complex tasks;
 - iii. Provide strategic planning for long term initiatives such as energy efficiency improvements and life cycle management of critical infrastructure; and
 - iv. Provide strategy to optimize and grow NCCIPS capabilities to attract new Customers.
- b. Facility Manager: The Facility Manager shall be designated by the Contractor and shall have complete authority to act for the Contractor on a day-to-day basis at the work site. The Facility Manager's physical location shall be at the NCCIPS facility. The Facility Manager shall possess at a minimum five years of recent experience in directing operations and maintenance of data center facility infrastructure for Tier III, multi-tenant, government data centers or equivalent experience. The Facility Manager shall be able to:
 - v. Understand the complete scope of all data center systems, processes, and infrastructure;
 - vi. Analyze complex project requirements;
 - vii. Build and lead a team with the skills to implement all data center maintenance and project requirements;
 - viii. Analyze the continuing performance of all aspects of the data center infrastructure and operations;
 - ix. Detect data center performance problem areas and suggest solutions; and
 - x. Serve as the NCCIPS representative to the Stennis Space Center Facilities Managers team.

The Contractor shall not replace any personnel designated as Key Personnel without the written concurrence of the CO. This notification shall be no later than ten (10) calendar days in advance of any proposed substitution and shall include justification including resume(s) of proposed substitution(s) in sufficient detail to permit evaluation of suitability (DRD 2.25-1).

NASA desires that critical skills be retained to insure continuity of performance for this critical facility. The Contractor shall retain appropriate levels of the incumbent workforce to ensure the intellectual knowledge is not jeopardized. In this context, critical skills retention is defined as retention of current personnel with a high level of intellectual knowledge of the NCCIPS facility including mechanical/electrical structures to ensure NCCIPS Customers experience no down time during and/or after the contract transition.

2.26 CONTINGENCY SUPPORT

In the event of a planned IT outage, equipment failure, hurricane, or declared disaster, a crew consisting of essential personnel shall be required to ensure continuous 24x7x365 operation. The makeup of that crew is defined in PWS section 2.8.1. Contingency support shall be approved in advance by the COR, CO, or other authorized Government approving official.

2.27 PHASE-IN & PHASE-OUT SERVICES

Phase-In: Following contract award, there shall be a sixty (60) day phase-in period, during which the Contractor shall ensure a smooth transition of services from the predecessor contract to the current contract.

Within ten (10) days after contract award, the Contractor and NSSC shall schedule and attend a Kick-Off Meeting among the COR, Program Manager, and cognizant NSSC personnel at the NSSC, Stennis Space Center, MS. The cost for phase-in services shall include all expenses (e.g., travel, labor, per diem) associated with the phase-in services, including travel to Stennis Space Center, MS for meetings. Meetings will be documented by the Contractor. Minutes shall include the following information: type/title of meeting and meeting date; purpose; location; summary of the discussions, decisions, agreements, and directions; list of attendees; and copies of action item sheets for any actions identified during the meeting. The Government will review and approve the meeting minutes and provide comments to the Contractor within fifteen (15) days of receipt. Meeting minutes are due NLT five (5) working days after the Kick-Off Meeting. Revisions shall be submitted, as required, ten (10) days after receipt of comments.

Phase-Out: Before performance on this Contract ends completely, the Contractor shall use reasonable efforts to facilitate a successful transition of services from it to the awardee under a subsequent contract for the same or similar requirement. The Contractor shall remain responsible for the effort covered by the PWS during phase-out activities. The Contractor shall develop with Government input, a plan of action to facilitate the transition of services to the incoming Contractor with no degradation of services.

3.0 FACILITIES MANAGEMENT

The purpose of PWS Section 3 is to identify requirements for the Contractor to provide facility program management, project support, operations, maintenance, engineering, network support, physical security, information security, inventory management, and shipping and receiving support. In addition, the Contractor shall provide maintenance and enhancements, when approved by the Government, of critical facilities infrastructure and IT hardware, software, and/or firmware in support of the NCCIPS 24x7x365 Tier III equivalent infrastructure for all NCCIPS facilities and for the telecommunications building as shown in TE-1 and TE-2. Specifically, the Contractor shall provide the following:

- a. Sensitive Compartmented Information Facilities (SCIF) support per Intelligence Community Standard 705-1 “Physical and Technical Security Standards for Sensitive Compartmented Information Facilities;”
- b. Energy management per the DCOI guidelines;
- c. Life cycle maintenance and 10-year replacement management of critical infrastructure per certified Original Equipment Manufacturer (OEM) recommendations, engineering equipment condition assessment, and industry best practices;
- d. Critical infrastructure management per Presidential Policy Directive PPD-21 “Critical Infrastructure Security and Resilience” and the most recent edition of “The Risk Management Process for Federal Facilities: An Interagency Security Committee Standard;”
- e. Maintenance and operations management per NPR 8831-2F “Facilities Maintenance and Operations Management;”
- f. IT systems integrations;
- g. Technical writing support, photography, and video as needed for the creation and maintenance of critical facilities documentation and internal training;
- h. Technical management support for NCCIPS projects either internally generated or Customer requested; identify resource requirements and recommend solutions, including plans, cost estimates, and schedules; coordinate authorization to proceed with the Government; and
- i. Project management per the following guidelines:
 - i. Ensure Project Managers are certified Project Management Professionals and shall be certified in Information Technology Infrastructure Library Foundation;
 - ii. Provide industry best practices in the management of projects;
 - iii. Develop project implementation plans and schedules to meet Customer requirements;
 - iv. Provide analysis and determination of electrical loads, heat loads, and fail over (redundancy) capacity to accommodate new or relocated IT systems;
 - v. Ensure costs and schedules developed are accurate and executed accordingly; and
 - vi. Ensure all internal and Customer Configuration Management procedures are followed during execution of Project Implementation Plans.

3.1 FACILITIES OPERATIONS AND MAINTENANCE

Facilities Operations and Maintenance includes design, implementation, operation and maintenance of a secure physical infrastructure for the NCCIPS data center environment. This includes:

- a. Power;

- b. HVAC;
- c. Cable management as it relates to reliability, expansion, and growth;
- d. Maintain training in Cardiopulmonary Resuscitation, Automated External Defibrillator, Electrical Room safety, and use of the fire extinguishers in accordance with applicable NFPA 70E and OSHA 29 Code of Federal Regulations (CFR) requirements including but not limited to 29 CFR 1910.303(a), 29 CFR 1910.303(d), 29 CFR 1910.303(g)(1), and 29 CFR 1910.303(e); and
- e. Be trained in the use of and be able to operate the Fire Extinguisher FE-25 discharge “abort” in all zones as well as the Master Abort located in the Facility Monitoring room.

3.1.1 Facilities Operations

Facilities Operations shall constantly monitor the NCCIPS facility to ensure that it remains operational, safe, and secure and that facility problems are quickly identified and resolved. Phone communications and radio communications shall be monitored continually to report on the status of NCCIPS and the SSC telecommunications building. Facilities Operations shall support SCIFs per Intelligence Community Standard 705-1.

Facilities Operations shall also:

- a. Conduct routine rounds to inspect all equipment and discover any issues before they become critical and to assess the conditions at NCCIPS according to documented procedures and checklists;
- b. Operate and monitor NCCIPS HVAC and power distribution systems in accordance with documented procedures and processes;
- c. Provide escort for un-cleared personnel;
- d. Provide incident reporting in accordance with DRD 3.1-1 in the event of equipment failure that impacts Customer environments or when requested by NASA;
- e. Maintain and update the Facilities Log as required;
- f. Maintain accurate daily logs using the NCCIPS SharePoint workflow system;
- g. Track inventory through the Facilities Log and Computerized Maintenance Management System (CMMS);
- h. Direct the development of required reports concerning Facilities Operations and ensure reporting accuracy;
- i. Monitor facilities to ensure continuous, uninterrupted power and cooling. These systems include, but are not limited to a SCADA system, a BAS, control, display and networking systems for the operation of the Power Control Room (PCR) and network monitoring equipment for Continuous Diagnostics and Mitigation (CDM) implementations;
- j. Continually monitor NCCIPS facility and SSC telecommunications building via installed cameras for signs of fire, smoke, equipment malfunction or security incident, and notify the Contractor Security POC and Facility Operations Supervisor, as appropriate;
- k. Provide “second person,” as needed, to Facility Operations staff responding to a facility emergency or malfunction in compliance with two-man rule per NCCIPS Safety Instructions;
- l. Assist Facility Operations lead in evacuation of NCCIPS facility in case of emergency and conduct roll call at designated muster site(s), if required;
- m. Reset fire alarm panels during emergencies, if directed by Fire Department or Facility Operations;

- n. Staff the camera monitoring room 24x7x365 with no less than two people. Ensure monitoring includes recording of videos and maintenance and retrieval of video archives for up to 60 days;
- o. Monitor Facility Access Control IT system and update data to ensure only authorized personnel are granted access to facility areas and computing zones. As needed, provide access reports as specified by NCCIPS management;
- p. Operate Facility Access Control Badging system to collect required data and generate NCCIPS badges for facility access. Ensure Access Control system displays proper credentials for each issued badge. Ensure NCCIPS permanent and visitor badges are appropriately controlled, issued, and logged. Ensure that all personnel permitted onsite have a current approved Site Visit Request or appropriate Security Clearance (permanent personnel). Ensure that all personnel onsite have up-to-date door access based upon monthly supervisory written approval. Ensure compliance with Physical Security and Facility Operations procedures and notify on-site Security or NCCIPS Security Coordinator immediately in case of any security incident or question. Properly secure data center zones, if required (e.g., X09 locks, alarms, Intrusion Detection Systems (IDS), etc.), as directed. If violations occur, the Contractor shall investigate, create and deliver incident reports to NCCIPS management and the NCCIPS Security Coordinator (DRD 3.1-2);
- q. Maintain a control log of all visitors who enter the main NCCIPS facility (Building 9325) and those requesting access to the various zones (Buildings 9323 and 9324). Direct visitors to the appropriate areas of the building being visited and provide escort as required;
- r. Ensure applicable NCCIPS WIs and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List;
- s. Assist in planning and establishing policies for managing physical security;
- t. Assist Security personnel with operation and oversight of the NCCIPS campus; and
- u. Maintain an Operations Call Desk function to receive and manage Trouble Calls. These Trouble Calls shall be prioritized in accordance with the work priorities below:
 - i. Emergency – Any work necessary to correct failures, incidents or events that constitute a danger to personnel or property; a violation of health, safety, and/or environmental regulations; or a significant impact to the NASA or tenant mission.
 - ii. Urgent – Work necessary to ensure continuous operations of a facility or system and/or to restore a facility or system to a healthful environment. The condition does not pose an immediate danger to personnel, property, or the NASA mission. However, if not responded to, the condition could result in an emergency situation.
 - iii. Routine – Facilities maintenance work that can be scheduled routinely within the capability of the organization. Contractor shall complete in order of receipt, subject to availability of resources, and consolidate by facility or zone or as directed to obtain efficiency of operation.
 - iv. Discretionary – Work that is desired but not essential to protect, preserve, or restore facilities and equipment; typically, new work that is not tied to a specific mission milestone.
 - v. Deferred – Work that may be safely, operationally, and economically postponed. The work should be done, but cannot be scheduled because of higher priority work, funds shortage, work site access, or conditions outside the control of the organization.

vi. Response and repair times for Trouble Calls

Work Priority	Notification	Execution
Emergency	Provide notification per existing operating procedures and/or work instructions and to the COR within thirty (30) minutes of failure and provide regular updates until repair are completed.	Respond immediately. Contractor shall be at the job site and working within thirty (30) minutes of failure. Work shall continue without interruption and shall arrest the emergency condition before departing the job site. Maximum repair time: twenty-four (24) hours.
Urgent	Provide notification per existing operating procedures and/or work instructions and to the COR within thirty (30) minutes of failure and provide regular updates until repair are completed.	Respond within twenty-four (24) hours. Maximum Repair Time: two (2) calendar days.
Routine	None required	Maximum Repair Time: five (5) calendar days.

3.1.1.1 Shipping & Receiving

The Contractor shall provide support for a dedicated shipping and receiving facility in accordance with NPR 4100.1 "NASA Supply Support and Material Management Procedural Requirement," as amended. Accurate database and tracking is required. The Contractor shall provide the required support for NCCIPS and the SSC telecommunications building including but not limited to:

- Receive shipments of IT and other parts, equipment, and supplies;
- Inspect and report any damage per DRD 2.18-3;
- Log assets into an asset tracking data base;
- Deliver select assets to NCCIPS Customers;
- Ensure applicable NCCIPS WIs and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List;
- Maintain a chain of custody as assets are transitioned to the various NCCIPS Customers; and
- Provide certified forklift operators as needed.

3.1.1.2 Custodial and Recycling

The Contractor shall be responsible for specialized cleaning of the approximately 86,000 square feet of raised floor areas in the NCCIPS data center (Buildings 9323 and 9324) and in the SSC telecommunications building data center. Custodial personnel are required to have a SECRET clearance. Special techniques required to maintain the integrity of the data center, include but are not limited to:

- High Efficiency Particulate Air (99.97% or higher for sub-micrometer particles) vacuum cleaning of the computer racks at least weekly;

- b. Vacuum computer room raised floor areas at least weekly;
- c. Damp mop of the computer room raised floor areas at least weekly;
- d. Cleaning of areas underneath the raised floor; and
- e. Dust mitigation and glass cleaning.

Additionally, the Contractor shall be responsible for providing custodial and recycling services, including glass cleaning, in all facilities identified by the use of “NCCIPS” in the Custodial and Recycling columns of TE-1 as shown. All areas require removal of trash and recyclables twice weekly.

In all areas, the Contractor shall provide response for spills, glass breakage, overflows, and blood-borne pathogens.

3.1.1.3 General Materials & Supplies

In accordance with PWS Section 2.12, the Contractor shall procure all materials and supplies required for establishment and replenishment of store stock and consumable items such as, but not limited to, light bulbs, office supplies, cleaning supplies, personal safety equipment, minor electrical supplies, and hand tools. These costs will be invoiced and reimbursed under the General Materials & Supplies CLINs.

3.1.1.4 Inspections

The Contractor shall perform box, package, and trash inspections for materials leaving data center zones, 7 days per week x 12 hours per day, 7:00am to 7:00pm Central. NCCIPS Operations shall dispatch a Facility Operator to inspect the materials prior to their introduction to or removal from the data center during the hours specified. After the specified hours, NCCIPS Operations shall dispatch the SSC roving armed security guard to inspect the materials prior to their introduction to or removal from the data center. In the event the SSC security guard is not available after the specified hours, NCCIPS Customers shall be directed to either wait for the SSC security guard or bring the materials to the NCCIPS Front Desk where one of the two Camera Operators in NCCIPS Access Control shall inspect the materials.

The Contractor shall also perform box, package, and bag inspections for NCCIPS Visitors at the NCCIPS Front Desk, 5 days per week (Monday – Friday) x 10 hours per day, 7:00am to 5:00pm Central. After the specified hours, the SSC security guard shall perform the box, package, and bag inspections. In the event the SSC security guard is not available after the specified hours, one of the two Camera Operators in NCCIPS Access Control shall inspect the box, package, or bag.

NCCIPS Customers will be directed to either wait for the patrol guard or bring the materials to the NCCIPS Front Desk, where one of the two designated Camera Operators in NCCIPS Access Control will inspect the materials. Note NCCIPS Front Desk package inspections shall only be performed for NCCIPS Visitors that will be bringing materials into the data center. NCCIPS-badged personnel are not routinely inspected at the Front Desk, but are subject to other random inspections.

3.1.2 Facilities Maintenance

The Contractor shall provide ongoing maintenance for items listed in, but not limited to, Attachment J-24, Government Furnished Property. Group A Property/Equipment shall be maintained by the Contractor through the utilization of certified OEM maintainers whether sourced internally to the Contractor or sourced via external maintenance subcontracts by certified OEM vendors. Group B Property/Equipment shall be maintained by the Contractor via preventative maintenance (no maintenance contract required). If Group B maintenance requirements exceed Contractor capabilities, an outside vendor may be requested to perform repairs. Group C Property/Equipment will be supported under NASA IT Services Program contracts and OEM warranties where applicable.

Facilities Maintenance shall constantly monitor the NCCIPS facility to ensure that it remains well maintained and that facility problems are quickly identified and resolved. Facilities Maintenance shall develop, implement, and maintain a preventive maintenance (PM) schedule, conduct corrective maintenance (CM), and support all associated operations services to ensure 24x7x365 business continuity of the NCCIPS data center support equipment. Routine PM schedules and CM conducted shall be reported in accordance with DRD 3.1-3.

TE-1, Facility Support Matrix, categorizes the NCCIPS facilities into Core Operations Facilities and Ancillary Support Facilities and indicates support required by either the NCCIPS Contractor or the SSC Contractor. According to TE-1, the NCCIPS Contractor shall provide most of the maintenance support for Core Operations Facilities with the SSC Contractor providing the balance of the maintenance support. For the Ancillary Support Facilities, the NCCIPS Contractor shall provide some maintenance support with the SSC Contractor providing most of the maintenance support. The Government will facilitate discussions during Phase-In to ensure that the NCCIPS Contractor and the SSC Contractor coordinate, establish, and document lines of demarcation for TE-1 maintenance support for both Core Operations Facilities and Ancillary Support Facilities.

Areas for PM and CM shall include but not be limited to:

- a. Electrical Distribution;
- b. UPS;
- c. Batteries;
- d. Chillers;
- e. Mechanical Infrastructure;
- f. Air Handlers;
- g. Environmental Management System;
- h. Fire Alarm and Suppression Systems;
- i. Access Control and Security Systems;
- j. Lightning Protection Systems; and
- k. Generators.

Facilities Maintenance shall also:

- l. Schedule vendor maintenance;
- m. Schedule and provide escort for, assistance to, and oversight of vendors during CM/PM work;

- n. Communicate to Facilities Engineering via the NCCIPS Work Request (WR) system and monthly to verify preventive maintenance;
- o. Perform routine diesel fuel (generator farm) testing, management, polishing, and reporting;
- p. Maintain master PM schedule;
- q. Implement and maintain scheduled maintenance using CMMS to automate day-to-day work tasks and reporting requirements. CMMS works in conjunction with the NCCIPS WR system to:
 - i. Maintain a database of all facility equipment to include life cycle management, estimated replacement cost, planned replacement date, and recommended OEM replacement date. Equipment replacement plans shall be produced annually, or as requested by NASA, in accordance with DRD 3.1-4; and
 - ii. Schedule routine maintenance;
- r. Validate readings on facility equipment UPS, Power Distribution Units, Chillers, CRAC units, etc.;
 - i. Schedule preventative maintenance;
 - ii. Generate work orders;
 - iii. Maintain equipment history; and
 - iv. Track warranty information to ensure continued service once the warranty expires.
- s. Coordinate all maintenance and infrastructure changes through a Configuration Control Board before maintenance is performed; and
- t. Ensure all lightning protection is adequate and maintained according to industry best practices.

3.1.2.1 General Facility / Grounds Maintenance

NCCIPS anticipates possible support being extended to replace general facility and grounds maintenance provided by Stennis Space Center. Should this occur, the Contractor may be tasked to provide such services. These services shall be negotiated and incorporated into the contract via a modification.

3.2 ELECTRICAL OPERATIONS

The Contractor shall provide ongoing electrical support for the NCCIPS Tier III equivalent data center, the SSC telecommunications building, and associated facilities. The Contractor shall fully document the NCCIPS PCR in accordance with DRD 3.2-1. All Contractor electricians shall be certified and shall follow applicable sections of 29 CFR 1910 "Occupational Safety and Health Standards," NFPA 70 National Electrical Code Standards, and NASA safety requirements in the execution of their duties, which include, but are not limited to:

- a. Facility Operations and Maintenance electrical support;
- b. Systems integration electrical support;
- c. Supporting Engineering with facility electrical enhancement/augmentation efforts;
- d. Electrical load analysis to support systems integrations;
- e. Installation and commissioning of Power Distribution Units;
- f. Fabrication of electrical wiring assemblies;
- g. Maintaining accurate electrical panel schedules;

- h. Ensuring applicable NCCIPS WIs and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List;
- i. Perform thermal imaging of critical electrical infrastructure as necessary; and
- j. Provide 24x7x365 facility support by certified electricians.

3.3 ENGINEERING

The Contractor shall provide facilities engineering analysis and support services for the entire NCCIPS campus and for the telecommunications building to include the following:

- a. Perform electrical and thermal load analysis to support Customer system integration;
- b. Ensure safe work practices meet or exceed NFPA70E including arc flash and arc blast analysis and safety implementations;
- c. Perform assessments as needed to support equipment installation and operation;
- d. (DEMAND) Provide engineering support for site preparation of requirements for internal facility modifications, both NCCIPS internal projects and Customer projects. This entails writing construction and modification requirements documents and statements of work, Bill of Materials to be purchased, reviewing engineering design proposals, sub-contract technical and project management, and commissioning of NCCIPS facility modifications;
- e. Maintain current records and drawings of Customer leased and occupied square footage by type of space occupied including floor layouts, rack elevations, and other drawings and provide reporting in accordance with DRD 3.3-1;
- f. Provide engineering support for SCIF per Intelligence Community Standard 705-1;
- g. Maintain engineering documentation for NCCIPS as-built configurations and update NCCIPS WIs and corresponding process-related documents as applicable. See TE-3, WI Document List. This includes maintaining accurate representations of the facility infrastructure via Computer Aided Design database and associated drawings;
- h. Provide sustaining engineering support for assessment and management of the current electrical, mechanical, communications, security (badging and camera), fire/life safety, inventory management and IT / Information Assurance (IA) facilities network support systems; and
- i. Provide engineering support to facility support systems for preventive and corrective maintenance, including support and maintenance of documented procedures for electrical and mechanical facility support systems.

3.4 REPAIR AND ALTERATION

The Contractor shall provide project management and reporting for all repair and alteration projects. All project management and reporting effort shall be covered as part of CORE SERVICES under this contract. This work shall include all resources necessary to track, analyze, and report project status and costs, including Gantt chart project schedule when deemed appropriate by the Government or the Contractor, with the level of project management intensity proportional to the complexity of the project in accordance with DRD 3.4-1. This requirement for project management shall be applicable to all facility projects, regardless of scope and contractor/agency. Operations and Maintenance projects exceeding \$150K and all Customer DEMAND projects shall be included in the DRD monthly reporting. Operations and Maintenance projects below \$150K implementation cost shall be tracked on electronic spreadsheet, and shall be available for Government review upon request.

For work outside of CORE SERVICES, project management specifically excludes responsibility for actual Field Engineering, Construction Management, Engineering/Design quality oversight, and Commissioning. However, the Contractor shall provide tracking and reporting for these work elements. For projects funded directly outside this contract, the Contractor shall continue to provide support for reporting, planning, scheduling, and implementation of facilities projects.

The Contractor shall also provide support for Customer and Government DEMAND projects. The source of funding for this labor support shall be charged according to the funding source of the project being supported. This support shall include:

- a. Providing escorts and access for planning and inspection associated with Projects or Engineering;
- b. Assisting with location and copies of configuration drawings, and facility equipment documentation, manuals and information;
- c. Repurposing or reconfiguration of existing data center and office spaces to support new Customer initiatives;
- d. Installing, maintaining and cutting raised floor tiles to accommodate systems integrations;
- e. Performing minor repairs to NCCIPS facility including walls, drop ceiling, minor utility relocations. This may require a third party vendor as approved by NASA;
- f. Providing general support for systems installations including the removal of debris generated from these installations;
- g. Cleaning and repairing facility infrastructure as required; and
- h. Providing certified forklift and aerial man lift operators as needed.

3.5 ENERGY MANAGEMENT

The Contractor shall comply with the energy efficiency and water conservation goals and requirements of the National Energy Conservation Policy Act 42 United States Code Chapter 91 Sections 8251-8287 as amended by the Energy Policy Act of 2005 and of Executive Order 13123 “Greening the Government through Efficient Energy Management,” dated June 3, 1999. The Contractor shall also comply with:

- a. NPR 8570.1 “NASA Energy Management Program” dated July 12, 2013 and
- b. Presidential Memorandum on Implementation of Energy Savings Projects and Performance-Based Contracting for Energy Savings, dated December 2, 2011.

The Contractor shall track and report the Power Usage Effectiveness (PUE) used to determine the energy efficiency of the data center. With the goal of sustaining a PUE of no more than 1.5 for the entire data center facility, the Contractor shall consider energy efficiencies when replacing equipment and identify other means of obtaining energy savings. Per Federal DCOI guidelines, the contract shall provide at least one certified Data Center Energy Practitioner to manage energy performance of NCCIPS, ensure best energy efficiency practices to improve PUE, and produce monthly reports (DRD 3.5-1).

The Contractor shall provide Computational Fluid Dynamics (CFD) modeling support as needed to ensure adherence to ASHRAE Thermal Guidelines for Data Processing Environments, 4th Edition, for data centers and efficient PUE Energy Management. The Contractor shall:

- c. Perform field verification of assets (rack elevations and assets) and respond to ad hoc Customer requests for reports;
- d. Ensure applicable NCCIPS WIs and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List; and
- e. Provide CFD and 3D modeling support as needed to ensure adherence to ASHRAE Thermal Guidelines for Data Processing Environments, 4th Edition, and efficient PUE Energy Management per American National Standards Institute / ASHRAE Standard 90.4-2016 “Energy Standard for Data Centers.”

3.6 INFORMATION TECHNOLOGY

The Contractor shall provide Facility IT, IA, system security plans, and IT Security Management Plan in accordance with DRD 2.7-1. The Contractor shall provide continuous monitoring reports as defined in DRD 3.6-1. Additionally, the Contractor shall provide, maintain, and update system administration and Industrial Control System support for the NCCIPS Facility to include the following:

- a. Register and maintain all NCCIPS applications/systems/websites in the NASA Agency Application registration system (e.g., System for Tracking and Registering Applications and Web sites / Applications Portfolio Assessment Tool);
- b. Develop and maintain an accurate and up-to-date as-built baseline Enterprise Architecture with updates to the baseline completed no later than 10 workdays days after IT project or initiative completion that results in changes to the Enterprise Architecture (DRD 3.6-2 and DRD 3.6-3);
- c. Support IT systems integration efforts (internal network cabling, patch panels, rack elevations, etc.);
- d. Provide hardware and application software maintenance (configuration, patches, updates, backups, scanning, etc.) for all NCCIPS facility infrastructure related IT equipment;
- e. Maintain accurate records documenting all physical layer communications connections throughout the NCCIPS campus and telecommunications building, including optical fiber and copper cabling runs, devices, and circuit patching information;
- f. Provide engineering, installation, and maintenance of NCCIPS campus local area networks;
- g. Assist in troubleshooting network connectivity and wide area communications issues for NCCIPS Customer and NCCIPS internal operations which may include interacting with the commercial circuit provider(s) supporting the specific NCCIPS Customer;
- h. Provide measurement and analysis of traffic volume data pertaining to implementation and support of the NCCIPS central site complex;
- i. Coordinate internal network configuration naming conventions and Internet Protocol addressing procedures with the host activity campus network infrastructure management organization to ensure consistency;
- j. Coordinate enforcement of security and access policies for the internal network with the host activity campus network infrastructure management organization per System Security Plan requirements;
- k. Comply with National Institute of Standards and Technology Special Publication 800-82 “Guide to Industrial Control System (ICS) Security”, as amended, in the management and maintenance of NCCIPS SCADA, BAS, and PCR building equipment controls/monitoring including code development, maintenance, updates, and patches;

- l. Ensure applicable NCCIPS WIs and corresponding process-related documents are updated as required to support changes in operational processes and NASA/Federal requirements. See TE-3, WI Document List;
- m. Ensure that NCCIPS network and computer security is in compliance with NPR 2810.1 “Security of Information Technology,” as amended;
- n. Support CDM requirements reporting per DRD 3.6-1 in accordance with OMB Circular A-130 “Management of Federal Information Resources” to:
 - i. Increase the overall security levels of the facility, reduce the number of breaches, reduce vulnerabilities and increase Federal Information Security Modernization Act of 2014 compliance scores;
 - ii. Provide better visibility of security posture and situational awareness at the department/agency level and the overall facility level; and
 - iii. Facilitate a continuous security improvement focus, with the ultimate goal of completely assessing a network within a 72-hour window.
- o. Support VOIP telephone system updates as needed in support of 24x7x365 VOIP operations. For billing purposes, maintain accurate records of phones as assigned to specific users;
- p. Implement and track the status of all security bulletins;
- q. Ensure NCCIPS assets comply with Federal and Agency IT requirements and policies;
- r. Assist in the preparation of Assessment and Authorization packages in accordance with NPR 2810.1;
- s. Generate and resolve any Plan of Action and Milestones in response to any security assessments or security findings (DRD 3.6-1);
- t. If a security incident occurs, the Contractor shall immediately inform NSSC Chief Information Security Officer and the COR. The Contractor shall participate in notifying the NASA Security Operations Center, collecting IDS, firewall, system and application logs and performing forensic analysis. The Contractor shall perform IT Security forensics and incident response and reporting per NSSC and Agency policy;
- u. Provide ad hoc supplemental IT hardware and software troubleshooting and resolution support as necessary to sustain day-to-day internal business activities of the NCCIPS Contractor and CS staff (where not provided under a NASA contract for managed desktop support). This includes but is not limited to support for desktop hardware, desktop operating systems, standard application software, email, Government Furnished Equipment printers, servers, and shared drives.; and
- v. Provide audio-visual (A/V) support to include ensuring proper setup, operation and maintenance of NCCIPS A/V systems and resolving A/V system problems with NCCIPS end users, external organizations and vendors. As required, the Contractor shall provide immediately available A/V support for special events and meetings. Audio-visual systems include projectors and public address systems.